

SUKKUR IBA UNIVERSITY
DEPARTMENT OF COMPUTER SYSTEMS ENGINEERING

FYDP Progress Report

Project Title:
SafeDrive AI: Intelligent dashboard device for driver anomaly detection

Muhammad Usman, 133-22-0016

Asma Channa, 133-22-0042

Abdul Moiz, 133-22-0032

Kashish Thorani, 133-22-0015

Supervisor

Dr. Abdul Sattar Chan

Sukkur IBA University

Supervisor

Engr. Umair Ayaz Kamangar

Sukkur IBA University

Progress Report Submission Date:

Details of FYDP Work:

Final Year Design Project Work	Hardware Based (%)	Software Based (%)
	60%	40%

	In term of (%)	Description
Executed Work	33%	• Applied foundational data augmentation techniques to increase dataset variability and effectively address class imbalance. • Trained the YOLOv8n-cls model using multiple hyperparameter configurations, including variations in batch size, learning rate, and training epochs, to optimize classification performance. • Observed that the YOLOv8-cls model delivered reliable and higher-accuracy results based on the conducted experiments. • Completed the full environment setup on the Raspberry Pi for on-device inference. • Deployed and tested the trained YOLOv8-cls model on the Raspberry Pi using a sample video to validate operational performance.
Remaining Work	67%	• Introduce an additional target class, “Drowsy” to the existing dataset, extending the current class taxonomy. • Apply advanced data augmentation techniques to enhance model robustness under challenging scenarios such as variable lighting and environmental fluctuations. • Retrain the YOLOv8-cls model using the expanded dataset and validate the updated performance metrics. • Convert the retrained model to ONNX format to enable deployment on the Raspberry Pi. • Perform model quantization to meet real-time inference requirements on constrained hardware. • Deploy the optimized model on the Raspberry Pi alongside the output-logic Python script. • Integrate a camera module with the Raspberry Pi and configure the output-logic script as a system service to ensure automatic execution at device startup.

Comments by Supervisor/Co-Supervisor:

Is the FYDP Project/Thesis ready for Evaluation I/II/III/final presentation? No ☐ Yes ☐

Signature

Supervisor
